Automatic Cattle Identification based on Multi-Channel LBP on Muzzle Images

Worapan Kusakunniran, Thanatchon Chaiviroonjaroen
Machine Vision and Information Transfer (MVIT) Research Group,
Mahidol Bremen Medical Informatics Research Unit (MIRU),
Faculty of Information and Communication Technology,
Mahidol University, Nakhon Pathom, Thailand
Email: worapan.kun@mahidol.edu, pop28843@gmail.com

***Abstract*—Every individual is unique. Biometrics authentication is mainly used to distinguish and identify each person. Parts of human body which are widely used for identification are fingerprints, faces and iris. In recent years, biometrics authentication begins to be used on animals, for the sake of population control, legal ownership and trade, and disease surveillance. This paper focuses on the automatic identification of cattles. Similar to human's fingerprint in term of uniqueness, cattle's muzzle is used in the identification process. In the conventional way, plastic ear tags are used to identify individual cattles. However, they can be worn down or lost easily. In addition, microchips are also used and implanted into cattles. This could injure them or cause some sickness. It is also expensive and requires human experts for the implant process. This paper introduces a novel solution using biometric images for the cattle identification. The proposed method extracts features from muzzle images using histogram of multi-channel Local Binary Pattern (LBP). This feature extraction is processed on sub-images to preserve the local spatial information of the muzzle patterns. Then, Support Vector Machine (SVM) is employed as the main classifier. The proposed method is evaluated using the published dataset containing 31 different cattles. It achieves the perfect performance of 100% accuracy.**

***Index Terms*—Cattle identification, Muzzle prints, LBP, SVM**

I. Introduction

Muzzle print or nose print, was investigated as the distinguished pattern for cattles since 1922. Those muzzles have uneven surface and contains countless varying arrangement and distribution of ridges and valleys, in which they are responsible for the formation of pattern on the muzzle. Muzzle print is used as a form of permanent identification because it is consistent over time and individualistic like human fingerprints.

Using the muzzle print for identification is challenging as the images are not always clear and conveniently recognizable. There are several methods proposed to improve the identification accuracy over years, either modifying images to be more clean or improving the identification algorithms. The first step was often removing noise on images by using filtering techniques such as average filter [1], median filter [1], and morphology filter [2].

A modified fuzzy filter, named Fuzzy noise filter, was proposed in 2016 and proved to have the great performance in removing noises [3]. After applying the filter, their features were extracted and stored for the later used. These features were unique to each individual cattle. In 2014, Gabor feature extraction is used to extract texture features from gray-scaled images. However, it is not effective against scaled or rotated images [3].

In later experiments, scale-invariant feature transform (SIFT) was used instead, as it was robust against rotated or shifted images. The random sample consensus (RANSAC) was also used to remove inaccurate SIFT out, for the better identification [4]. Later experiment used speed-up robust features (SURF), as it was claimed to be better than SIFT [5]. Local binary patterns (LBP) was another popular method used to extract features, along with linear discriminant analysis (LDA) which was used to keep only important features [6].

Lastly, the classifier was applied for the final identification. Decision tree is one of the earlier method used, which gave fair results [1]. While SVM, KNN, Nearest Neighbor, and Naive Bayes Classifiers were used in Tarek Gaber's experiment. It showed that SVM gave the best accuracy [6]. Gaussian-based SVM classifier achieved the best accuracy compared to other SVM kernels [7]. Recently, the paper in [8] applied SURF for the feature extraction and employed SVM as the classification tool. In 2016, artificial neural network (ANN) [9] was attempted for the classification of muzzle images. Then, in [10], Gaussian pyramid, SURF and LBP were combined for the feature extraction.

With the main objective of enhancing the muzzle image-based identification performance, this paper proposes the method based on the histogram of multi-channel LBP, PCA, and SVM. The multi-channel LBP with various setting could capture multiple primitive texture patterns of the muzzle image.

The proposed method is evaluated using the dataset of 31 cattles with 7 muzzle images each [6] [7] [8] [11]. It achieves the perfect identification accuracy of 100%. The technical details are explained in the section II. Experimental results are reported in the section III. Conclusions are drawn in the section IV.

978-1-5386-7407-9/18/$31.00 ©2018 IEEE

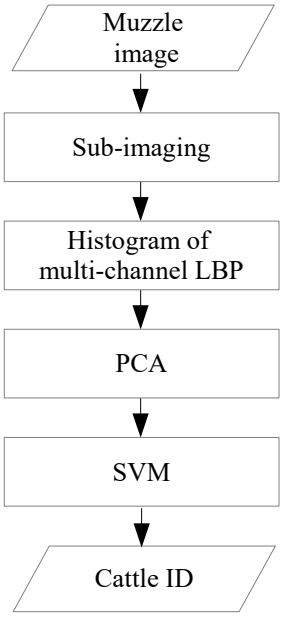

Fig. 1. The overview of proposed method.

II. Proposed Methods

The figure 1 shows the overview framework of the proposed method. Given the muzzle image, it is divided into sub-images. The later processes are applied on each sub-image, to preserve the local spatial information of the muzzle patterns. Then, the LBP with a specific setting is applied to extract primitive texture pattern. The LBP image with n-channel is constructed with n different sets of LBP parameters (i.e. radius and number of neighboring pixels). In the computed LBP image, each pixel contains n values, in which each value is generated from one setting of LBP.

Then, the histogram is constructed for each sub-image of the LBP image. All generated histograms are later concatenated. The PCA is applied to reduce the dimension of the extracted histogram feature. Finally, the SVM is trained on the extracted features, as the classification tool for identifying each individual cattles. The following sub-sections explain technical details of key techniques of LBP, PCA and SVM.

A. Local Binary Pattern

Local Binary Pattern (LBP) is a texture operator which labels each pixel on the image using the threshold of neighbor pixels around it, in the form of a binary number. If the value of the neighboring pixel is higher or equal to the value of the center pixel, then its LBP value is set to be one. Otherwise, it is set to be zero. This binary sequence is then converted into a decimal number for each pixel, as shown in the equation (1). Due to its high discriminative power and low computational complexity, LBP has been considered as one of the very popular descriptors. This process is applied on each sub-image of the given muzzle image.

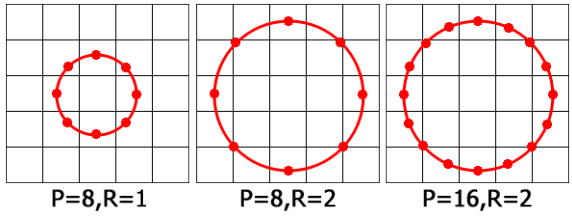

Fig. 2. Positions of the neighboring pixels in LBP, according to P and R.

$$LBP_{P,R} = \sum_{p=0}^{p=P-1} s(g_p - g_c)2^p \tag{1}$$

where P is the number of the neighboring pixels, R is the radius from the neighboring pixels to the center pixel (see Fig.2), g_c corresponds to the gray level of the center pixel, g_p denotes the gray level of the p equally spaced pixels on the circle of radius $R(R > 0)$, and s is the threshold function. The s(x) = 1 if x $\geq$ 0, otherwise 0.

All pixels in the image are contributed into the LBP code. With different P and R values, different primitive texture patterns could be captured. In this paper, the multi-channel LBP image is constructed using multiple setting of P and R. Then, the histogram of the LBP-image in each sub-area is generated. Later, all histograms of all sub-images are concatenated, representing the feature vector of the muzzle image in this stage.

B. Principal Components Analysis

Principal Component Analysis (PCA) is a technique to reduce the dimension of feature vectors, but still contains most of the information of the original vectors. It explores correlations between training samples across data variables. In this paper, the PCA is applied on the concatenated histogram, for reducing the feature dimension and noise.

First, the data matrix H is constructed as shown in the equation (2). In H, rows represent different data samples (i.e. muzzles) and columns represent different data values of the feature vector (i.e. concatenated histogram).

$$H = \{h_{i,j}\}_{N \times M} \tag{2}$$

where $h_{i,j}$ is a value at the jth position in the feature vector of the ith sample, N is the total number of samples, and M is the dimension of the feature vector. Then, the equation (3) is performed to standardize the matrix H.

$$\widetilde{H} = \{\frac{h_{i,j} - \mu_j}{\sigma_j}\}_{N \times M} \tag{3}$$

where μ_j and σ_j are the mean and standard deviation respectively of the data in column j of the matrix H. Then, the covariance matrix (C) can be constructed as below.

$$C = \frac{\widetilde{H}^T \widetilde{H}}{N} \tag{4}$$

where $\widetilde{H}^T$ is the matrix transpose of $\widetilde{H}$. The eigenvectors and their corresponding eigenvalues are computed by applying the eigen-decomposition on the matrix C. The k eigenvectors corresponding to their k largest eigenvalues are selected and used to form up the PCA projection matrix, as below.

$$k = \underset{1 \leq k \leq M}{\operatorname{argmin}} \frac{\sum_{i=1}^{k} |v_i|}{\sum_{i=1}^{M} |v_i|} > \frac{r}{100} \tag{5}$$

where v_i is the eigenvalue of the ith eigenvector, v_i is in an descending order, and r is pre-defined as the amount of variation coverage of the whole feature vectors. In this paper, $r = 99\%$.

Then, the PCA projection matrix is used to project the feature vector (i.e. concatenated histogram) with the dimension of M to k, while $k < M$.

C. Support Vector Machine

Support Vector Machine (SVM) is the technique to split a dataset into two independent subsets, using a hyperplane. Thus, SVM is to seek out the hyperplane that maximizes the margin between the nearest samples of the two subsets. The output of the linear SVM is shown in the equation (6).

$$u = \vec{w} \cdot \vec{x} - b \tag{6}$$

This is corresponding to the decision function as in the equation (7).

$$f(x) = sign((\vec{w} \cdot \vec{x}) + b) \tag{7}$$

where $w \in R^N$ is the normal vector to the hyperplane, $x \in R^N$ is the input vector, and $b \in R$ is a constant. That is, the SVM attempts to find a gap between the two classes by maximizing the distance between the nearest samples of the two classes. The points lying on the boundaries are called support vectors. The dataset must follow the below core condition. Each sample must belong to only one of the two classes of the dataset, as shown in the equation (8).

$$\{x_i, y_i\}, i = 1, \cdots N, x_i \in R^d, y_i \in \{-1, 1\} \tag{8}$$

III. Experiments

The proposed method is evaluated using the dataset of 31 cattles [6] [7] [8] [11]. There are 7 muzzle images of each cattle. The sample images are shown in Fig. 3. Several scenarios of the experiment are explored as in the following sub-sections.

A. LBP Selection

In this experiment, for each cattle, 4 muzzle images are used for training and the other 3 muzzle images are used for testing. The LBP feature with one setting of P and R is used, without the histogram and sub-imaging processes. Three types of LBP are explored, including original LBP (OLBP), extended LBP (ELBP), and variance-based LBP (VARLBP). The identification performance is shown in Table I. As can be seen in Table I, VARLBP with R=5 gives the highest accuracy.

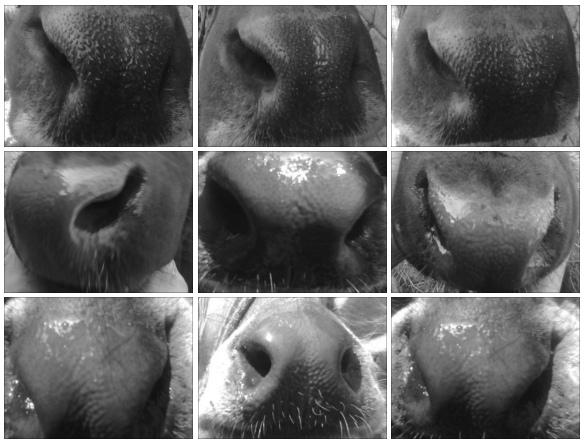

Fig. 3. The sample muzzle images. Rows represent different cattles. Columns represent different muzzle images of the same cattle.

B. Histogram of VARLBP

In this experiment, for each cattle, 4 muzzle images are used for training and the other 3 muzzle images are used for testing. Since the VARLBP with R=5 and P=16 gives the highest accuracy in the above scenario, this setting is used as the baseline in this scenario. Then, the histogram of VARLBP with R=5 and P=16, without the sub-imaging process is used in this experiment. The identification performance is shown in Table II.

This can be seen that the performance is dropped when compared with the LBP feature without the histogram processing. This is because when the LBP is re-organized into the histogram, the local spatial pattern information is missing. Thus, in the next scenario of the experiment, the sub-imaging process will be added for the feature extraction, in order to preserve the local information of the muzzle patterns.

C. Histogram of VARLBP on Sub-images

In this experiment, the same scenario as in the above subsection is employed but it is applied on the sub-images instead of the whole image. Then, the histograms of all sub-images are concatenated into a single feature vector. The identification performance is shown in Table III. It is shown that the histogram with 256 bins and 7×7 grid size is chosen over the others. This is because the more number of histogram's bins could be more robust and preserve more spatial information of the patterns.

D. Histogram of multi-channel VARLBP on Sub-images

In this scenario, the multi-channel LBP is introduced on top of the previous scenario. The identification performance is shown in Table IV. The result shows that the more numbers of LBP channels could lead to the better identification performance.

E. Increasing the number of training images

In the previous sub-sections, 4 muzzle images of each cattle are used for the training process. In this experiment, increasing

TABLE I
THE EXPERIMENTAL RESULTS FOR THE SCENARIO OF LBP SELECTION.

Types	R	P	Accuracy (%)
OLBP	-	-	15.05
ELBP	1	8	14.00
	1	16	16.13
	1	24	21.51
	1	32	19.35
	2	8	11.83
	2	16	12.90
	2	24	16.13
	2	32	18.28
	3	8	10.75
	3	16	25.81
	3	24	13.98
	4	16	30.11
	4	20	37.63
	4	24	41.94
	4	28	38.71
	4	32	13.98
	5	16	39.78
	5	24	45.16
	5	28	40.86
	5	32	46.24
	5	48	46.24
	6	16	38.71
	6	24	44.09
	6	32	41.94
	6	48	41.94
VARLBP	1	8	37.63
	2	8	40.86
	3	8	49.46
	4	8	53.76
	5	8	59.14
	5	16	60.22
	5	24	59.14
	6	8	60.22
	6	16	58.06
	6	24	59.14
	6	32	58.06
	7	8	56.99
	7	16	56.99
	7	24	55.91

TABLE II
THE EXPERIMENTAL RESULTS FOR THE SCENARIO OF HISTOGRAM OF VARLBP.

Number of bins	Accuracy (%)
16	47.31
32	51.61
64	48.39
128	54.84
256	52.69

the number of training images are employed to enhance the performance of the cattle identification. The identification performance is shown in Table V.

IV. CONCLUSION

This paper proposes a novel solution of using muzzle images to identify individual cattles. The histogram of multi-channel LBP on sub-images provides the best performance.

TABLE III
THE EXPERIMENTAL RESULTS FOR THE SCENARIO OF HISTOGRAM OF VARLBP ON SUB-IMAGES.

Number of bins	Image partitioning	Accuracy (%)
32	3×3	76.34
	4×4	79.57
	5×5	80.65
	6×6	79.57
	7×7	80.65
	8×8	78.49
	9×9	76.34
	10×10	79.57
64	3×3	76.34
	4×4	79.57
	5×5	79.57
	6×6	76.34
	7×7	77.42
	8×8	78.49
	9×9	75.27
	10×10	78.49
128	3×3	74.19
	4×4	78.49
	5×5	78.49
	6×6	80.65
	7×7	78.49
	8×8	78.49
	9×9	76.34
	10×10	75.27
256	3×3	75.27
	4×4	79.57
	5×5	79.57
	6×6	77.42
	7×7	80.65
	8×8	76.34
	9×9	77.42
	10×10	76.34

TABLE IV
THE EXPERIMENTAL RESULTS FOR THE SCENARIO OF HISTOGRAM OF MULTI-CHANNEL VARLBP ON SUB-IMAGES.

Number of LBP channels	Accuracy (%)
1	80.65
2	83.87
3	86.02

PCA is applied to reduce the feature dimension and SVM is used as the classification tool. With the 6 training images of each cattle, the proposed method achieves 100% accuracy. In the future work, it could be validated with the larger number of cattles. Also, other texture analysis techniques could be combined with the current used feature to form up the stronger classifier.

REFERENCES

[1] H. M. E. H. Ibrahim El-Henawy, Hazem. M. El bakry and N. Mastorakis, "Muzzle feature extraction based on gray level co-occurrence matrix," *International Journal of Veterinary Medicine*, vol. 1, pp. 16–24, 2016.

[2] H. A. Mahmoud and H. M. R. E. Hadad, "Automatic cattle muzzle print classification system using multiclass support vector machine," *International Journal of Image Mining*, vol. 1, no. 1, pp. 126–140, 2015.

TABLE V
THE EXPERIMENTAL RESULTS FOR THE SCENARIO OF HISTOGRAM OF MULTI-CHANNEL VARLBP ON SUB-IMAGES, WITH VARIOUS NUMBERS OF TRAINING IMAGES.

# of training images	# of LBP channels	Accuracy (%)
4	1	80.65
	2	83.87
	3	86.02
5	1	82.26
	2	90.32
	3	90.32
6	1	93.55
	2	100.00
	3	100.00

[3] D. M. G. Anusha Edwin, "Fuzzy noise extraction from muzzle images for cattle identification," *International Journal of All Research Education and Scientific Methods (IJARESM)*, vol. 4, pp. 367–371, 2016.

[4] H. M. Z. Ali Ismail Awad, Aboul Ella Hassanien, "A cattle identification approach using live captured muzzle print images," in *International Conference on Advanced Machine Learning Technologies and Applications*.

[5] A. M. A. Ary Noviyanto, "Automatic cattle identification based on muzzle photo using speed-up robust features approach," *Recent Advances in Information Science*, pp. 110–114.

[6] H. A. E. Alaa Tharwat, Tarek Gaber Aboul Ella Hassanien and M. F. Tolba, "Cattle identification using muzzle print images based on texture features approach," in *Proceedings of the Fifth International Conference on Innovations in Bio-Inspired Computing and Applications IBICA 2014*. Springer, 2014, pp. 217–227.

[7] A. E. H. Alaa Tharwat, Tarek Gaber, "Cattle identification based on muzzle images using gabor features and svm classifier," in *International Conference on Advanced Machine Learning Technologies and Applications*. Springer, 2014, pp. 236–247.

[8] S. Ahmed, T. Gaber, A. Tharwat, A. E. Hassanien, and V. Snáel, "Muzzle-based cattle identification using speed up robust feature approach," in *Intelligent Networking and Collaborative Systems (INCOS), 2015 International Conference on*. IEEE, 2015, pp. 99–104.

[9] I. El-Henawy, H. M. El Bakry, and H. M. El Hadad, "Cattle identification using segmentation-based fractal texture analysis and artificial neural networks," *International Journal of Electronics and Information Engineering*, vol. 4, no. 2, pp. 82–93, 2016.

[10] S. Kumar, S. K. Singh, and A. K. Singh, "Muzzle point pattern based techniques for individual cattle identification," *IET Image Processing*, vol. 11, no. 10, pp. 805–814, 2017.

[11] A. Tharwat, T. Gaber, and A. E. Hassanien, "Two biometric approaches for cattle identification based on features and classifiers fusion," *International Journal of Image Mining*, vol. 1, no. 4, pp. 342–365, 2015.